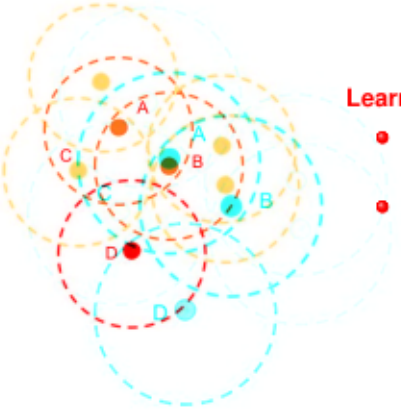
# Interpretable Machine Learning

## Increasing Trust in Explanations




**Learning goals**

- Understand the aspects that undermine users' trust in an explanation
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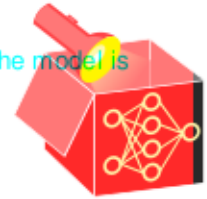



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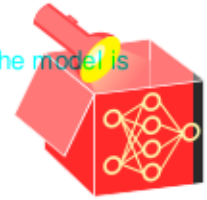



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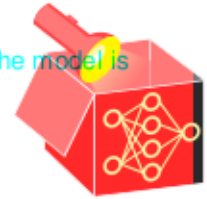

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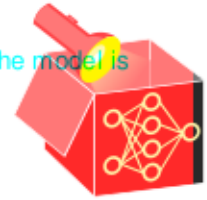

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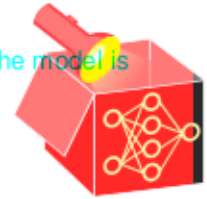

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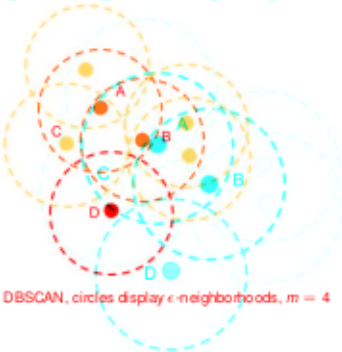


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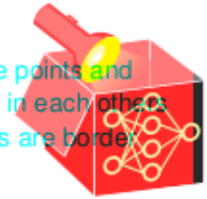



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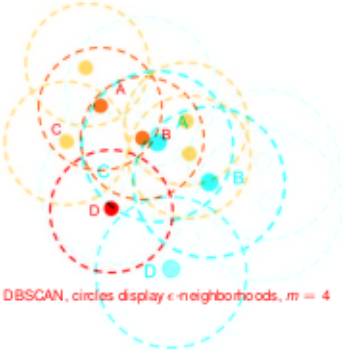


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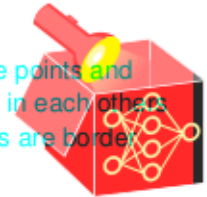



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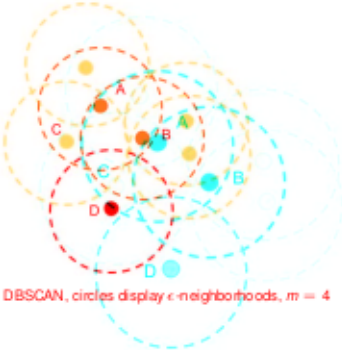


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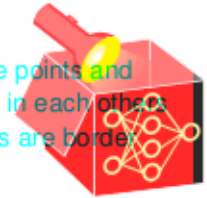



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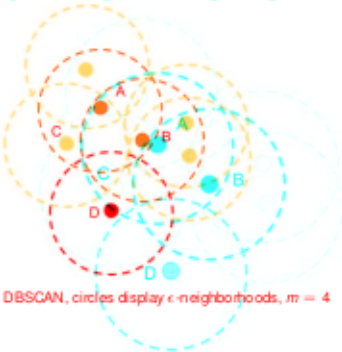


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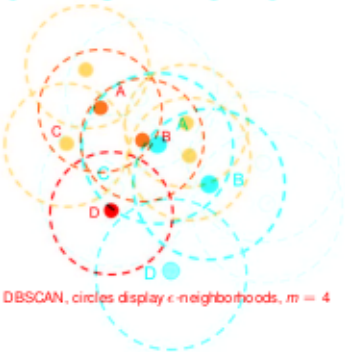


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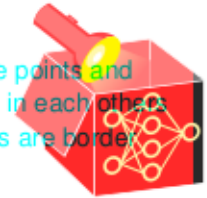

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- Differentiate between different kinds of uncertainty:
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     $\rightsquigarrow$ are ML models non-robust, e.g., because they are trained on noisy data?
- We focus on explanation uncertainty
  - Even with the same model and same (or similar) data points, we can receive different explanations

# ROBUSTNESS MEASURE FOR LIME AND SHAP

- Objective: Similar explanations for similar inputs (in a neighborhood)



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- For LIME and SHAP, notion of stability based on **locally Lipschitz continuity**

Alvarez-Melis and Jaakkola 2018 :

An explanation method $g : \mathcal{X} \to \mathbb{R}^m$ is locally Lipschitz if

- for every $\mathbf{x}_0 \in \mathcal{X}$ there exist $\delta > 0$ and $\omega \in \mathbb{R}$
- such that $\|\mathbf{x} - \mathbf{x}_0\| < \delta$ implies $\|g(\mathbf{x}) - g(\mathbf{x}_0)\| < \omega \|\mathbf{x} - \mathbf{x}_0\|$

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- According to this, we can quantify the robustness of explanation models in terms of $\omega$:
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- $\omega$ is rarely known a-priori but it could be estimated as follows:

$$\hat{\omega}_X(\mathbf{x}) \in \arg\max_{\mathbf{x}^{(i)} \in \mathcal{N}_\epsilon(\mathbf{x})} \frac{||g(\mathbf{x}) - g(\mathbf{x}^{(i)})||_2}{d(\mathbf{x}, \mathbf{x}^{(i)})},$$

where $\mathcal{N}_\epsilon(\mathbf{x})$ is the $\epsilon$-neighborhood of $\mathbf{x}$



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